INCH-POUND

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MILITARY SPECIFICATION

POLYPROPYLENE CORES, STRAND CENTERS, AND SUBSTRANDS FOR WIRE ROPE

This specifications is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for monofilament, multifilament, and slit or fibrillated fiber po]ypropylene cores, strand centers, and substrands.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these are those cited in the solicitation or contract.

FEDERAL

FED-STD-191 Textile Test Methods

(Copies of this document is available on-line at https://quicksearch.dla.mil)

Comments, suggestions, or questions on this document should be addressed to DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM International

ASTM D792	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1525	Standard Test Method for Vicat Softening Temperature of Plastics
ASTM D2257	Standard Test Method for Extractable Matter in Textiles

(Copies of these documents are available on-line at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The material used in cores, strand centers, and substrands shall be virgin polypropylene fiber containing sufficient heat stabilizers to meet the softening temperature requirements of 3.1.3 and sufficient ultraviolet (U.V.) stabilizer to meet the requirements 3.4.2.

3.1.1 Tenacity (strength). The material shall have tenacity (strength) of 3.6 grams to 4.8 grams per denier.

3.1.2 Specific gravity. The material shall have a specific gravity no greater than 0.91.

3.1.3 Softening temperature. The material shall have a softening temperature of at least 300 degrees Fahrenheit (°F). All fibers shall contain sufficient heat stabilizers to meet the softening temperature requirements when tested as specified in 4.2.

3.1.4 Material form. The material may consist of multifilaments, monofilaments, connected monofilament, or film, slit or fibrillated, as agreed between the wire rope producer and the core producer.

3.2 Filaments and film size. Filaments and film used to produce cores, strand centers, and substrands shall be the same size.

3.2.1 Multifilament size. Filaments used for multifilament shall range in size from 6 denier to 15 denier per filament.

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3.2.2 Monofilament size. Monofilament shall range from 100 to 800 denier per filament.

3.2.3 Film size. Slit and fibrillated fiber shall be composed of tape whose size shall be a maximum of 75,000 denier per filament.

3.3 Non-fibrous material. No extraneous material shall be added for the purpose of weighting the cores, centers, or substrands, except coloring pigments.

3.3.1 Moisture content. The moisture content shall not exceed 1.0 percent of the dry weight of the core, strand center, or substrand.

3.3.2 Lubricant. The cores, strand centers, and substrands shall be lubricated with a lubricant specified by the wire rope producer except the lubricant shall not contain asphaltic material or animal fats. The lubricant content shall be 4 ± 2 percent when tested as specified in 4.1.3.

3.4 Resistance and aging. Finished cores, strand centers, and substrands shall meet the following requirements.

3.4.1 Heat aging resistance. Finished cores, strand centers, and substrands shall be resistant to aging. They shall have a breaking strength loss no greater than 10 percent when compared with the original specimens when tested as specified in 4.1.1.

3.4.2 Ultraviolet resistance. Substrands for spring lay rope shall be resistant to ultraviolet degradation. The specimens removed from the substrands shall have a breaking strength loss of no greater than 50 percent when compared with the original specimens when tested as specified in 4.1.2.

3.5 Construction. The cores, strand centers, and substrands shall consist of a laid construction usually using 3 strands. When specified, 4 strands may be used with a core (sec 6.2). Either construction shall be a hard lay capable of supporting the wire rope and acceptable to the wire rope producer.

3.6 Identification. The producer of the polypropylene cores, centers, or substrands shall, if required (see 6.2), identify his product by inserting a paper or colored polyoefin monofilament marker within one strand.

3.7 Color. Unless otherwise specified (see 6.2) the color of the cores and centers shall be charcoal grey or black, while the color of the substrands shall be black.

3.8 Workmanship. Cores, centers and substrands shall be evenly laid. The individual strands in the cores, centers, and substrands shall be free from kinks, strand knots, and darting yarns or bulged strands.

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4. VERIFICATION

4.1 Conformance testing. The methods of testing as specified in table I shall be followed. The physical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods.

TABLE I. Test methods.

Characteristic	Requirement Reference	Test Method	Tests per Sample
Material	3.1	Certification	1
Tenacity (strength)	3.1.1	FED-STD-191 method 4100[1, 2]	3
Specific gravity	3.1.2	ASTM D792	3
Softening temperature	3.1.3	ASTM D1525	1
Filament size	3.2	Certification	1
Moisture content	3.3.1	FED-STD-191 method 2600[1]	1
Lubricant	3.3.2	Certification for type of lubricant 4.1.3 for lubricant quantity	1 2
Heat aging	3.4.1	4.1.1, FED-STD-191 method 4100[1, 2]	3
Ultraviolet resistance	3.4.2	4.1.2, FED-STD-191 methods 4100 and 5804[1, 2]	3
Construction	3.5	Visual	1
Identification marker	3.6	Visual	1
Color	3.7	Visual	1
Workmanship	3.8	Visual	1

1 – Tests to determine conformance with requirements may be made under prevailing atmospheric conditions except in settlement or dispute in which case the tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.
2 – Tenacity results may be used as basis of Heat aging and Ultraviolet resistance strength loss.

4.1.1 Heat aging test. Three specimens shall be heated in a convection air oven at 175 ± 2 °F for *5* days. These specimens shall then be removed from the oven and conditioned for 24 hours in a standard atmosphere for textiles prior to being tested for strength. The percent change in breaking strength shall be calculated based on the average found for three unexposed specimens as specified in method 4100 of FED-STD-191.

4.1.2 Ultraviolet resistance test. Three specimens shall be exposed for 500 hours in the weathering device described in method 5804 of FED-STD-191. The percent change in breaking strength shall be calculated on the average found for three unexposed specimens as specified in method 4100 of FED-STD-191.

4.1.3 Lubricant quantity test**.** An extraction procedure in accordance with the Soxhlet apparatus option of ASTM D2257 and using chloroform as the solvent, shall be used to determine the amount of lubricant applied to the material. The quantity of lubricant is per 3.3.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to

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ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The polypropylene cores, strand centers, and substrands covered by this specification are intended for use in wire rope.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Type of laid construction required (see 3.5)
(c) Manufactures identification, if required (see 3.6)
(d) Color, if other than specified (sec 3.7)

6.3 Subject term (key word) listing.

Monofilament
Multifilament
Slit film
Fibrillated film

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Navy – SH
DLA – GS

Preparing Activity:
DLA – GS5

Review Activities:
DLA – CC

(Project 4020-2020-001)

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